

6G Weekly Survey — Jun 08 - Jun 14, 2026

18 items across 6 topics, ranked by importance.

Gen AI based Channel Modeling

Implementation and Calibration of 3GPP-Compliant ISAC Channel Simulator

Chien-Han Wu, Ming-Chun Lee, Ta-Sung Lee · arXiv eess.SP · 2026-06-05

★★★★■ Provides essential infrastructure for 6G ISAC research by delivering a validated, reproducible implementation of the official 3GPP channel model, directly enabling standardization-aligned development and testing. This work implements the 3GPP TR 38.901 ISAC channel model simulator and performs comprehensive calibration against reference results from 3GPP member companies. The authors address implementation ambiguities in the specification by documenting key implementation details and verifying consistency with standardized test cases. They release the simulator as open-source on GitHub with datasets and calibration results to enable reproducible ISAC research.

Contribution: First fully documented, calibrated, and open-source implementation of the 3GPP standardized ISAC channel model with explicit handling of specification ambiguities.

Relevance to your work: Directly relevant for your ISAC research—provides a calibrated simulation platform for evaluating AI-native physical layer designs and learned sensing-communication receivers against 3GPP-compliant channel conditions.

Double-Directional Wireless Channel Modeling Using Statistics-Aided Machine Learning

Richmond Boamah, Ferdous Pervej · arXiv cs.IT · 2026-06-04

★★★★■ This tackles a fundamental limitation in ML-based channel modeling (variable MPC counts) with a principled solution validated on both synthetic and deterministic datasets, directly applicable to 6G system-level simulation. This paper addresses double-directional wireless channel modeling by proposing a statistics-aided ML approach that handles variable numbers of multipath components across time. The method selects a fixed subset of top-M MPCs and uses a hybrid TimesNet-TimeFilter architecture trained to match channel statistics rather than exact realizations. Validation on both stochastic and ray-tracing datasets shows improved performance over baselines for long-term channel prediction.

Contribution: The statistics-aided training strategy enables fixed-dimension ML models to generate statistically consistent channel realizations despite variable MPC counts, avoiding the typical requirement for exact future channel alignment.

Relevance to your work: Directly relevant to your generative channel modeling work, offering a complementary approach to GAN-based methods by focusing on statistical consistency for double-directional massive MIMO scenarios.

ISAC (sensing + communication)

Foundation Models for Wireless Communications: From PHY Intelligence to Network Autonomy

Le Liang, Jiajia Guo, Jun Zhang, Chan-Byoung Chae, et al. · arXiv eess.SP · 2026-06-04

★★★★■ Strong conceptual framework from credible researchers that synthesizes an emerging trend and provides taxonomic clarity for applying foundation models to 6G physical layer and network problems, though lacks concrete performance benchmarks.

This paper surveys the application of foundation models (large-scale pre-trained AI models) to 6G wireless systems across three paradigms: adapting off-the-shelf models to wireless tasks, building wireless-native

foundation models trained on radio data, and agentic models that enable autonomous network orchestration. The authors examine applications to physical layer processing, resource management, ISAC, advanced MIMO, and semantic communications. The work argues foundation models can address 6G's complexity through zero-shot generalization and few-shot adaptation rather than task-specific training.

Contribution: First comprehensive survey categorizing foundation model approaches for wireless into three progressive paradigms (adapted, wireless-native, agentic) and mapping their application to 6G technologies including ISAC and semantic communications.

Relevance to your work: Directly relevant as it addresses AI-native air interface design through foundation models, with explicit coverage of ISAC, semantic communications, and learned physical layer processing aligned with your Beyond-5G research portfolio.

Rate-Splitting--Inspired Uplink Near-Field ISAC

Anup Mishra, Israel Leyva-Mayorga, Petar Popovski · arXiv eess.SP · 2026-06-05

★★★★■ Strong theoretical contribution addressing fundamental receiver design for uplink ISAC with near-field modeling, from credible authors; novel rate-splitting framework with rigorous large-array analysis relevant to 6G massive MIMO-ISAC systems.

This paper applies rate-splitting principles to uplink near-field ISAC, splitting communication messages across the sensing operation to manage inter-functionality interference. Closed-form expressions for communication-rate and sensing-rate are derived under residual sensing interference, with the RS-inspired achievable region matching or strictly enlarging the NOMA-inspired time-sharing region. Large-array analysis using aperture-aware near-field models reveals finite rate limits as array size grows.

Contribution: First application of rate-splitting to uplink ISAC that generalizes sensing-centric and communication-centric endpoint designs, proving the RS boundary can strictly enlarge the sensing-communication tradeoff region beyond time-sharing.

Relevance to your work: Directly relevant to your ISAC research: advances receiver architectures for managing sensing-communication interference in uplink, with near-field massive MIMO analysis critical for sub-THz 6G systems where near-field effects dominate.

SkySense: A Semi-Supervised Generative Framework for UAV Localization in ISAC Networks

Shenghan Luo, Yin Xu, Jie Yang, Yang Wang, et al. · arXiv eess.SP · 2026-06-02

★★★★■ Demonstrates practical ISAC sensing capability (sub-10 cm UAV localization) with minimal supervision and real-world validation, addressing a critical barrier to deploying learning-based positioning in commercial systems where labeled training data is prohibitively expensive.

This paper addresses CSI-based UAV localization in ISAC networks under severe labeled-data scarcity and multipath ambiguity. The authors propose a semi-supervised framework combining a self-supervised encoder that learns from unlabeled CSI trajectory sequences with a consistency model (diffusion derivative) that maps latent features to coordinates using minimal labeled data. Real-world measurements show 9.77 cm localization error with only 1% labeled data and 1-2 step inference, outperforming fully supervised baselines.

Contribution: First application of consistency models to CSI fingerprinting localization, enabling diffusion-like generative modeling with 1-2 step inference instead of hundreds, while addressing mean collapse through conditional distribution modeling under extreme label scarcity.

Relevance to your work: Directly relevant to your ISAC research: combines CSI-based sensing with modern generative models, addresses the label-efficiency problem endemic to learning PHY methods, and validates on real measurements rather than simulations—methodology could extend to beam management or channel prediction under sparse feedback.

Robust Secure Beamforming for Movable Antenna Enhanced Integrated Sensing and Communications

Yuan Chen, Ning Wei, Ahmad Bazzi, Xiangyu Dong, et al. · arXiv eess.SP · 2026-06-05

★★★★ Solid incremental work combining three emerging topics (ISAC, movable antennas, physical layer security) with competent optimization methodology, though movable antenna technology remains speculative for near-term deployment.

This work addresses robust beamforming design for movable antenna (MA)-enhanced ISAC systems under imperfect eavesdropper CSI. The authors jointly optimize transmit beamforming and physical antenna placement to maximize radar SINR while maintaining secure communication rates. A BCD-based algorithm combining successive convex approximation and fractional programming solves the challenging non-convex problem with coupled antenna position and channel coefficient dependencies.

Contribution: First integration of movable antenna technology with secure ISAC beamforming, introducing robust optimization under eavesdropper CSI uncertainty alongside the novel design space of optimizable antenna positions.

Relevance to your work: Directly relevant to your ISAC research agenda, particularly for understanding how reconfigurable antenna architectures (beyond RIS) can jointly optimize sensing and communication performance under practical security constraints.

3D Spherical Fluid Antennas for Spatially Reconfigurable Communications

Wenchi Cheng, Hongyun Jin, Zhuohui Yao · arXiv eess.SP · 2026-06-04

★★★★ Solid conceptual extension of fluid antennas to 3D with clear system design and applications, though feasibility of hardware implementation at mmWave/sub-THz frequencies and performance comparison to conventional massive MIMO with digital beamforming remains unclear.

This paper proposes a 3D spherical fluid antenna system (3D SFAS) that extends fluid antenna concepts from 1D/2D apertures to full 3D spatial reconfiguration by dynamically activating radiating elements across different spherical regions. The architecture enables joint array-level region switching and element-level aperture/topology reconfiguration for flexible beamforming, multi-region transmission, and blockage adaptation. Numerical results demonstrate performance gains through this spatial reconfigurability in various communication scenarios.

Contribution: First proposal of a 3D spherical fluid antenna architecture that combines region-level and element-level reconfiguration for comprehensive spatial adaptability in 3D propagation environments, moving beyond the 2D position switching studied in prior FAS work.

Relevance to your work: Directly relevant to ISAC applications where joint spatial reconfiguration could enable adaptive sensing apertures and blockage-resilient communication, and potentially applicable to NTN scenarios requiring omnidirectional coverage and dynamic beam steering for LEO satellite handovers.

X-Band UAV-enabled Integrated Sensing and Communications for Vehicular Networks

Remon Polus, Soumaya Cherkaoui · arXiv cs.IT · 2026-06-03

★★★★ Solid incremental work applying ISAC principles to UAV-vehicular context with realistic channel models, but optimization approach and X-band focus are fairly conventional for the domain.

This paper optimizes time-division between sensing and communication for a UAV operating at X-band serving vehicular networks, accounting for UAV altitude constraints and realistic fading (single/double-shadowing models). The authors formulate a joint optimization that ensures minimum communication rate guarantees while maintaining sensing reliability. Simulations show how channel quality and target distance drive adaptive time allocation between ISAC functions.

Contribution: Provides a practical time-allocation framework for UAV-based ISAC under X-band propagation constraints specific to vehicular scenarios, explicitly modeling air-to-ground shadowing effects.

Relevance to your work: Directly relevant as an ISAC use case combining aerial platforms with vehicular sensing, though the time-allocation trade-off is a well-studied aspect of ISAC resource management rather than advancing physical-layer integration or AI-native techniques.

SNF-PRP: A Covert Integrating Sensing and Communications Framework

Dhrumil Bhatt, Vidushi Kumar · arXiv eess.SP · 2026-06-02

★★★★ Solid theoretical contribution addressing a genuine security concern in ISAC with rigorous analysis and 5G numerology validation, though practical deployment challenges and adversary model limitations remain unexplored. This paper addresses the vulnerability that sensing signals in ISAC systems can reveal both transmitted data and the act of sensing to passive adversaries. The authors propose sub-noise-floor pseudo-random probing (SNF-PRP) for OFDM-based ISAC that operates below the noise floor with ϵ -covertness guarantees via KL divergence analysis, exploiting spreading gain across subcarriers. Simulations in 5G NR n78 demonstrate sub-0.5m range and velocity accuracy at -12 to -15 dB probing power while maintaining covertness $5.8\times$ below detection thresholds.

Contribution: First framework providing formal covertness guarantees for ISAC sensing signals operating below the noise floor with closed-form derivation of integration length requirements to achieve target Cramér-Rao bounds.

Relevance to your work: Directly relevant to your ISAC research interests, particularly addressing the tension between sensing performance and physical-layer security that will be critical for 6G JCAS standardization and military/dual-use applications.

Instantaneous Risk Minimization for Secure Integrated Sensing and Communication

Chao Ge, Na Zhao, Yuan Shen · arXiv eess.SP · 2026-06-02

★★★★ Solid incremental work addressing practical security considerations in ISAC with a principled optimization framework, but limited novelty in algorithmic techniques and lacks experimental validation or real-world channel modeling.

This paper develops a robust beamforming framework for secure ISAC systems that minimizes instantaneous information leakage risk rather than maximizing ergodic secrecy rate. A hierarchical iterative algorithm addresses the non-convex multi-objective optimization by refining uncertainty regions in an outer loop while optimizing beamforming in an inner loop. The method provides per-transmission physical layer security guarantees for ISAC systems under worst-case channel uncertainty.

Contribution: Novel risk-minimization approach to ISAC security that provides per-transmission guarantees rather than average secrecy metrics, with a tractable hierarchical algorithm coupling sensing performance to security region refinement.

Relevance to your work: Directly relevant to your ISAC research interests, particularly for understanding security-sensing trade-offs in joint systems where beamforming must simultaneously serve communication, sensing, and anti-eavesdropping objectives.

The Preisach Extremum Stack is a Shannon-Minimal Sufficient Statistic for Rate-Independent Functionals

Piotr Frydrych · arXiv cs.IT · 2026-06-03

★★★ Solid theoretical work on hysteresis modeling with potential applications in material science and control systems, but the connection to wireless communication or sensing is unclear and the result appears domain-specific. The paper proves that the Preisach extremum stack is a Shannon-minimal sufficient statistic for all rate-independent (time-reparametrization-invariant) functionals, meaning it captures all information needed for such queries with minimum mutual information. The characterization theorem shows any computable rate-independent functional can be expressed as a function of this stack, enabling online estimation of Preisach measures with $O(k \cdot L^2)$ memory per time step. The result provides an information-theoretic foundation for hysteresis modeling in rate-independent systems.

Contribution: First rigorous proof that the Preisach extremum stack achieves Shannon-minimality for the entire class of rate-independent functionals, establishing fundamental limits on data compression for hysteretic systems.

Relevance to your work: Tangential—while ISAC involves sensing and estimation theory, this work addresses classical hysteresis in rate-independent systems (magnetic materials, plasticity) rather than

channel sensing, radar, or wireless signal processing.

6G Physical Layer

A Cancellation Mechanism in AFDM Radar Sensing: Exact Fisher Information and Delay-Doppler Decoupling

Tingjun Lyu, Yunmei Shi · arXiv eess.SP · 2026-06-03

★★★★■ Strong theoretical contribution from a credible group that analytically characterizes AFDM sensing performance and proves fundamental advantages over OFDM for ISAC, directly addressing a 6G waveform candidate with validated results at realistic vehicular and LEO parameters.

The paper derives closed-form Fisher information and Cramér-Rao bounds for delay-Doppler estimation in AFDM radar, revealing that AFDM's chirp-periodic prefix exactly cancels frequency drift from chirp modulation. This decoupling mechanism makes AFDM provably less delay-Doppler-coupled than OFDM, with delay estimation accuracy improving quadratically with chirp rate while Doppler estimation remains unaffected. The framework reduces to classical OFDM radar theory as chirp rate vanishes.

Contribution: First closed-form Fisher information analysis for AFDM radar that identifies the chirp-periodic prefix as the structural element enabling delay-Doppler decoupling, providing analytic proof of superiority over OFDM rather than numerical bounds.

Relevance to your work: Directly relevant to your work on 6G waveforms (AFDM) and ISAC, providing rigorous theoretical foundation for AFDM's sensing capabilities and analytic design insights for chirp-rate optimization in joint communication-sensing systems.

Adaptive c_{chirp} -Perturbed AFDM Waveform Design for Integrated Sensing and Communication

Shiqi Cui, Fan Zhang, Yuanshuo Gang, Zeping Sui, et al. · arXiv eess.SP · 2026-06-03

★★★★■ Addresses two critical practical limitations of AFDM (high PAPR and sensing ambiguity) with a computationally feasible solution from a credible waveform research group, advancing AFDM toward real 6G ISAC deployment.

This paper optimizes the pre-chirp parameter c_{chirp} in affine frequency division multiplexing (AFDM) waveforms for ISAC systems to simultaneously reduce peak-to-average power ratio (PAPR) and improve autocorrelation sidelobe levels for sensing. A data-driven, real-time optimization framework uses a spectral projected-gradient algorithm with closed-form gradients, achieving receiver-comparable complexity. Simulations show improved sensing-communication trade-offs and better BER under power amplifier nonlinearity.

Contribution: First work to adaptively optimize AFDM's c_{chirp} parameter in real-time for joint PAPR reduction and autocorrelation sidelobe suppression in ISAC, using an efficient gradient-based algorithm suitable for practical implementation.

Relevance to your work: Directly addresses 6G waveform design (AFDM) and ISAC optimization, combining your interests in physical-layer waveforms and integrated sensing-communication with practical implementation constraints for next-generation air interfaces.

NTN / satellite

Dynamic FDD for Spectrum Sharing in Non-Terrestrial Networks

Sourav Mukherjee, Bho Matthiesen, Armin Dekorsy, Petar Popovski · arXiv cs.IT · 2026-06-03

★★★★■ Addresses a critical interference problem unique to dense NTN deployments with a novel spectrum flexibility approach, highly relevant as mega-constellations scale toward 6G integration.

This paper addresses inter-satellite interference in dense LEO mega-constellations by proposing dynamic reassignment of FDD bands instead of fixed uplink/downlink allocations. The authors formulate a joint

optimization of band assignment, user scheduling, and power allocation, solving the mixed-integer non-convex problem via equivalence transforms, alternating optimization, and commercial solvers. Numerical results show up to 30% throughput improvement over conventional static FDD in dense deployments.

Contribution: First study to propose and optimize dynamic FDD band reassignment specifically for managing inter-satellite interference in LEO mega-constellations, departing from fixed band allocations.

Relevance to your work: Directly relevant to your NTN research focus, particularly for LEO and direct-to-cell systems where interference management at scale is a fundamental challenge for 6G non-terrestrial architectures.

MediaTek demonstrates live NR-NTN LEO satellite connection at Wireless Japan 2026

MediaTek Blog · 2026-06-08

★★★★■ Significant commercial validation of NTN standards implementation, but this is a demonstration rather than novel research—confirms technology maturity for 6G non-terrestrial network integration.

MediaTek demonstrated a live 3GPP NR-NTN connection to a LEO satellite constellation at Wireless Japan 2026, showcasing direct-to-device satellite connectivity using standards-compliant Release 17/18 protocols. The demonstration likely validates end-to-end integration of chipset-level NTN support with commercial LEO infrastructure. This represents a commercial milestone rather than a research contribution, confirming industry readiness for satellite-cellular convergence.

Contribution: First public live demonstration of MediaTek's chipset supporting 3GPP NR-NTN with operational LEO satellites, validating commercial viability of standards-based satellite direct-to-device connectivity.

Relevance to your work: Directly relevant as empirical validation of NTN deployment feasibility; informs practical constraints for AI-native PHY design in satellite scenarios including propagation delay, Doppler, and power limitations.

AI-native air interface

WiSER: A Wireless Scene Encoder for Geometry-Grounded Multi-View Wireless Prediction

Jing Qiao, Yiyang Guo, Hao Ye · arXiv eess.SP · 2026-06-03

★★★★■ Strong contribution establishing geometry-conditioned multi-task propagation prediction with public ray-tracing pipeline, directly relevant to AI-native physical layer design and site-specific digital twins for 6G network planning.

WiSER introduces a unified neural architecture that maps sparse 3D indoor scene geometry plus transmitter location into a shared scene representation, from which two specialized decoders predict both dense radiomaps and sparse multipath channel impulse responses. The model is trained on a co-registered dataset pipeline using ScanNet++ indoor scenes and Sionna ray tracing to produce aligned voxel inputs, radiomap labels, and CIR tap sets. Experiments show WiSER outperforms single-task baselines for both radiomap coverage and multipath delay-power prediction.

Contribution: First architecture to jointly predict aggregate radiomap coverage and structured multipath CIR taps from a shared transmitter-conditioned 3D scene encoding, demonstrating that geometry-grounded representations can serve multiple heterogeneous propagation prediction tasks.

Relevance to your work: Directly relevant: provides foundation-model building blocks for learned PHY channel modeling, integrates geometry into neural channel representations, and addresses multi-view propagation queries needed for AI-native air interfaces and digital-twin-enabled network planning.

Environment-Division Multiple Access: an Enabler for AI-Native Multiple Access

Zhiguo Ding · arXiv cs.IT · 2026-06-05

★★★★ Solid conceptual framing that organizes emerging ideas (RIS, AI-native air interfaces) under a unified multiple access taxonomy, though specific technical contributions or performance gains are not demonstrated.

The paper proposes Environment-Division Multiple Access (EDMA), a new multiple access paradigm that exploits wireless propagation environment features, flexible antennas, and proactive environment reconfiguration (e.g., RIS) to separate users. It distinguishes AI-assisted EDMA (using AI to optimize existing schemes) from AI-native EDMA (end-to-end learning replacing conventional designs). The work discusses how EDMA complements traditional orthogonal and non-orthogonal schemes by adding an environment dimension to user separation.

Contribution: Introduces EDMA as a conceptual framework that positions environment reconfiguration and advanced antenna systems as a distinct multiple access dimension, bridging RIS, beamforming, and AI-native PHY design.

Relevance to your work: Directly relevant: articulates how AI-native PHY design (your core interest) can reshape multiple access by treating the propagation environment as an exploitable resource rather than a constraint.

Semantic communication

Adapting Diffusion Language Models for Lossless Pixel-Level Image Transmission

Tianqi Ren, Rongpeng Li, Xianfu Chen, Yingyu Li, et al. · arXiv cs.IT · 2026-06-04

★★★★ Solid incremental work applying diffusion models to a well-defined problem with clear experimental validation, though the focus on exact pixel recovery is narrow and less aligned with semantic communication's core value proposition of graceful degradation.

This paper proposes a discrete diffusion model for lossless pixel-level image transmission over noisy channels, replacing raster-order autoregressive coding with bidirectional masked denoising that restores multiple tokens per step. The framework combines a diffusion-based source codec with separate channel coding, introducing Halton-guided denoising order, mask-ratio-aware scheduling, and temperature calibration to bridge generation and arithmetic coding. Experiments on AWGN and Rayleigh fading channels demonstrate superior exact-recovery rates versus baseline semantic and lossless communication schemes.

Contribution: First adaptation of discrete diffusion language models to lossless image transmission, showing that progressive bidirectional restoration provides better source representations for noisy channels than sequential autoregressive coding.

Relevance to your work: Tangential—while semantic communication is a 6G theme, this work reverts to lossless transmission, and the diffusion-based source coding technique does not directly address physical-layer AI, channel modeling, or task-oriented communication objectives in your research portfolio.